

Structures at six: the Eisenstein geometry of E_6 , a parity obstruction for bilinear $\text{AME}(4, d)$ constructions, and the anatomy of two $\text{AME}(4, 6)$ representatives

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Abstract

Local dimension six carries two famous exceptions: there is no pair of orthogonal Latin squares of order six (Tarry), and the maximal number of mutually unbiased bases in $\mathbb{C}^6$ remains unknown. The first exception was resolved quantum-mechanically in 2022 by the construction of an absolutely maximally entangled state $\text{AME}(4, 6)$, equivalently a 2-unitary matrix of order 36. This note collects four independent results, all verified computationally from first principles.

(i) We give an explicit realization of the $\mathbb{Z}[\omega]$ -module structure of the E_6 root lattice through the fourth power of a Coxeter element, and verify that the resulting 12 complex lines in $\mathbb{C}^3$ decompose into exactly 4 orthogonal frames that are pairwise mutually unbiased. Thus E_6 realizes the *complete* MUB system of a qutrit, and the count of positive roots factors as $36 = 12 \times 3$.

(ii) We prove a parity obstruction: a flat 2-unitary of order d^2 whose phase is a bilinear form over $\mathbb{Z}/d$ exists *if and only if* d is odd. The proof is two lines and uses only that the units of $\mathbb{Z}/d$ are odd for even d . This explains why the finite-field construction available at every odd d has no analogue at $d = 6$, and shows the obstruction is one of parity rather than of non-prime-power order.

(iii) We reconstruct and independently verify two Butson-type $\text{AME}(4, 6)$ representatives with all phases in $\mu_6 = \mathbb{Z}[\omega]^\times$, and show that their phase function is not a polynomial of degree ≤ 2 over $\mathbb{Z}/6$, is not translation covariant, and does not factor through the Chinese Remainder decomposition $\mathbb{Z}/6 \cong \mathbb{Z}/2 \times \mathbb{Z}/3$.

(iv) We compute the automorphism group of the golden $\text{AME}(4, 6)$ representative and find it to be $\mathbb{Z}/2$ at the level of the modulus pattern, with the nontrivial element *failing* to lift to a monomial symmetry of the state. In particular the natural conjecture that the golden state is covariant for the exceptional action of $\text{PGL}_2(\mathbb{F}_5)$ on the six points of $\mathbb{P}^1(\mathbb{F}_5)$ is false for this representative. We show instead that φ enters *locally*, through the 2×2 block relations, as the icosahedral vertex direction $(1, \varphi)$.

(v) We record a companion arithmetic thread attached to the same modulus: a proof via the valence formula that the modular-zero polynomial in weight $12m$ has degree $m(m-1)$; an independent computation of the weight-36 Hecke cubic, whose discriminant shares the 12-digit prime 269461929553 with the sextic field of modular zeros; and unconditional numerics for $L(s, \chi_{-3})$, the L -function of the character underlying the Eisenstein master identity.

1 Introduction

Two exceptional phenomena meet at local dimension six.

The first is combinatorial. Euler conjectured in 1779, and Tarry proved in 1900, that no two orthogonal Latin squares of order six exist; 6 is the unique order > 2 for which this fails [1, 2, 3]. The best classical configuration attains 34 of the 36 officer pairs. In 2022 Rather et al. [4] showed that the problem admits a quantum solution: an absolutely maximally entangled state of four six-level systems, $\text{AME}(4, 6)$, equivalently a 2-unitary matrix of order 36, equivalently a perfect tensor with four legs of dimension six. The amplitudes of their solution involve the golden ratio $\varphi = (1 + \sqrt{5})/2$ and phases that are 20th roots of unity; it is known as the *golden* AME state. Subsequently it was shown that infinitely many inequivalent solutions exist [6], and two further constructions produced representatives that are Butson-type complex Hadamard matrices with all 1296 entries of equal modulus and phases in μ_6 [7, 8].

The second exception is measurement-theoretic. In prime-power dimension d there exist $d + 1$ mutually unbiased bases (MUBs), and this is maximal. In dimension six only three MUBs are known, and whether four exist is a long-standing open problem. Six is the smallest dimension where the question is open.

That the same dimension hosts both exceptions is folklore-level motivation for asking whether the objects that repair the first say anything about the second. This note is a report of what we could establish by direct computation, kept scrupulously separate from what we could not. Section 3 treats an Eisenstein structure in dimension three that turns out to be exactly a complete MUB system; Section 4 isolates the arithmetic obstruction at six and shows it is a parity phenomenon valid for all even d ; Section 5 dissects two $\text{AME}(4, 6)$ representatives. Appendix A tabulates the epistemic status of every claim.

All computations were performed independently of the cited literature: root systems were generated from Cartan matrices by breadth-first reflection closure, and the AME matrices were rebuilt from their defining data and re-verified, rather than assumed.

2 Preliminaries

Let $d \geq 2$ and write $N = d^2$. Index $\mathbb{C}^N \otimes \mathbb{C}^N$ by four indices, and for $X \in \mathbb{C}^{N \times N}$ define the *reshuffling* and *partial transpose*

$$X_{jk;lm}^{\text{R}} = X_{jl;km}, \quad X_{jk;lm}^{\text{T}} = X_{jm;lk}.$$

Definition 1. $U \in \mathbb{U}(d^2)$ is *2-unitary* if U , U^{R} and U^{T} are all unitary. Equivalently the associated four-index tensor $\psi_{jklm} = d^{-1}U_{jk;lm}$ is a *perfect tensor*: every reduction to two of the four legs is maximally mixed, $\text{tr } \rho^2 = d^{-2}$ for all six bipartitions. Equivalently $|\psi\rangle \in \text{AME}(4, d)$ [10].

Definition 2. H of order N is a *complex Hadamard matrix* if $HH^\dagger = N\mathbb{I}$ and $|H_{jk}| = 1$ for all j, k ; it is of *Butson type* $\text{BH}(N, q)$ if in addition every entry is a q -th root of unity [12].

We call a 2-unitary *flat* if all $d^2 \times d^2$ entries have modulus d^{-1} , i.e. if dU is a complex Hadamard matrix. It is known that 2-unitaries do not exist for $d = 2$ [11]; for $d \neq 2, 6$ they are furnished by orthogonal Latin squares as permutation matrices, and $d = 6$ is the case resolved in [4].

Throughout, $\omega = e^{2\pi i/3}$, $\mathbb{Z}[\omega]$ is the ring of Eisenstein integers, μ_k is the group of k -th roots of unity, and $\mathbb{Z}[\omega]^\times = \mu_6$.

3 The Eisenstein geometry of E_6 and the qutrit MUB system

The number 36 occurs twice in our story: as $|\Phi^+(E_6)|$ and as 6^2 . The following makes the first occurrence structurally transparent. That the E_6 root system carries a $\mathbb{Z}[\omega]$ -structure

related to the Hessian configuration of 12 lines is classical, arising from the theory of complex reflection groups; that the Hesse configuration encodes the qutrit MUB system is known in quantum information. Our contribution is an explicit, uniform realization through the Coxeter element together with complete verification of the resulting geometry.

Theorem 3. *Let Φ be the root system of E_6 with $W = W(E_6)$, Coxeter element c , and invariant bilinear form B normalized so that $B(r, r) = 2$ for roots. Put $g = c^4$. Then:*

- (a) $g^3 = 1$, g preserves B , and g is fixed-point-free: $\det(g - 1) = 27 \neq 0$.
- (b) $J := (2g + 1)/\sqrt{3}$ satisfies $J^2 = -1$ and $B(Ju, v) = -B(u, Jv)$, so $h(u, v) := B(u, v) + iB(u, Jv)$ is a Hermitian form making the ambient space $\mathbb{C}^3$ and the E_6 lattice a rank-3 $\mathbb{Z}[\omega]$ -module, with $h(r, r) = 2$ for roots.
- (c) The group $\mu_6 = \{\pm 1, \pm g, \pm g^2\}$ acts freely on the 72 roots, with exactly 12 orbits of size 6. Hence

$$72 = 12 \times 6 = \#\{\omega\text{-lines}\} \times |\mathbb{Z}[\omega]^\times|, \quad 36 = |\Phi^+| = 12 \times 3.$$

- (d) Let $L_1, \dots, L_{12} \subset \mathbb{C}^3$ be the resulting complex lines. Then for $i \neq j$ the quantity $|h(r_i, r_j)|^2$ takes exactly two values, 0 and $4/3$. Each line is orthogonal to exactly two others, so the 12 lines partition into 4 orthogonal frames of 3; and for lines in different frames the normalized overlap is

$$\frac{|h(r_i, r_j)|^2}{h(r_i, r_i)h(r_j, r_j)} = \frac{4/3}{4} = \frac{1}{3}.$$

Consequently the 12 lines form $d + 1 = 4$ pairwise mutually unbiased bases of $\mathbb{C}^3$: the complete MUB system of a qutrit.

Proof (computational). The 72 roots were generated from the E_6 Cartan matrix by closing the six simple roots under simple reflections; all have B -norm 2. The Coxeter element has order 12, consistent with $h(E_6) = 12$; since the exponents of E_6 are 1, 4, 5, 7, 8, 11, none is divisible by 3, so c^4 has no eigenvalue 1 and is fixed-point-free, which is what (a) records numerically as $\det(g - 1) = 27$. Items (b)–(d) were verified entrywise; the orthogonality graph on the 12 lines is 3-regular with components $\{0, 7, 11\}, \{1, 6, 10\}, \{2, 5, 8\}, \{3, 4, 9\}$ in the ordering produced by our enumeration, and all $\binom{12}{2} - 4\binom{3}{2} = 66 - 12 = 54$ cross-frame pairs give exactly $1/3$. $\square$

Remark 4. The identity $g^2 + g + 1 = 0$ on the ambient space is what allows $J = (2g + 1)/\sqrt{3}$ to square to -1 ; the factor $\sqrt{3}$ is the different of $\mathbb{Q}(\omega)$, and $\det(g - 1) = 27 = 3^3$ is the norm of $(1 - \omega)^3$ up to units. The chain A_1 ($h=2$), A_2 ($h=3$), G_2 ($h=6$), F_4 ($h=12$), E_8 ($h=30$) of root systems whose exponents are exactly the totatives of h was also verified; it places the mod-6 unit pattern $\{1, 5\}$ as the exponent set of G_2 .

4 A parity obstruction for bilinear 2-unitaries

At every odd d there is a one-line construction of a perfect tensor, $|\psi\rangle \propto \sum_{i,j} |i, j, i + j, i + 2j\rangle$ over $\mathbb{Z}/d$. Its Fourier dual is a flat 2-unitary whose phase is a bilinear form. The following theorem determines exactly when such an object exists, and shows the failure at $d = 6$ is a parity phenomenon.

For $M = \begin{pmatrix} m_{11} & m_{12} \\ m_{21} & m_{22} \end{pmatrix} \in M_2(\mathbb{Z}/d)$ and $\zeta = e^{2\pi i/d}$, define $U^{(M)} \in \mathbb{C}^{d^2 \times d^2}$ by

$$U_{(x,y),(u,v)}^{(M)} = \frac{1}{d} \zeta^{x(m_{11}u + m_{12}v) + y(m_{21}u + m_{22}v)}. \quad (1)$$

Lemma 5. $U^{(M)}$ is 2-unitary if and only if all three of

$$\det M = m_{11}m_{22} - m_{12}m_{21}, \quad m_{11}m_{22}, \quad m_{12}m_{21}$$

are units of $\mathbb{Z}/d$.

Proof. Write the exponent as $E = m_{11}xu + m_{12}xv + m_{21}yu + m_{22}yv$.

Unitarity of U . For row indices $(x, y), (x', y')$ with $(s, t) = (x - x', y - y')$,

$$(UU^\dagger)_{(x,y),(x',y')} = \frac{1}{d^2} \sum_{u,v} \zeta^{(sm_{11}+tm_{21})u} \zeta^{(sm_{12}+tm_{22})v},$$

which by orthogonality of characters vanishes unless $(s, t)M \equiv 0$, in which case it equals 1. Hence U is unitary iff $(s, t)M \equiv 0 \Rightarrow (s, t) = 0$, i.e. iff $\det M \in (\mathbb{Z}/d)^\times$.

Unitarity of U^R . Reading E with row index (x, u) and column index (y, v) , the terms $m_{11}xu$ and $m_{22}yv$ depend only on the row, respectively only on the column; they contribute diagonal unitary factors on either side and do not affect unitarity. The genuine pairing between row and column is $m_{12}xv + m_{21}yu$, of the form (1) with matrix $\begin{pmatrix} 0 & m_{12} \\ m_{21} & 0 \end{pmatrix}$, whose determinant is $-m_{12}m_{21}$. So U^R is unitary iff $m_{12}m_{21}$ is a unit.

Unitarity of U^Γ . Reading E with row index (x, v) and column index (u, y) , the terms $m_{12}xv$ and $m_{21}yu$ are row-only and column-only respectively, and the pairing matrix is $\text{diag}(m_{11}, m_{22})$ with determinant $m_{11}m_{22}$. So U^Γ is unitary iff $m_{11}m_{22}$ is a unit. $\square$

Theorem 6 (Parity obstruction). *A matrix $M \in M_2(\mathbb{Z}/d)$ with $U^{(M)}$ 2-unitary exists if and only if d is odd. For odd d one may take $M = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}$.*

Proof. Suppose d is even. Every unit of $\mathbb{Z}/d$ is odd. By Lemma 5 both $m_{11}m_{22}$ and $m_{12}m_{21}$ are units, hence odd; therefore $\det M = m_{11}m_{22} - m_{12}m_{21}$ is a difference of two odd numbers, hence even, hence not a unit of $\mathbb{Z}/d$. This contradicts Lemma 5.

If d is odd, take $M = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}$: then $m_{11}m_{22} = 2$, $m_{12}m_{21} = 1$ and $\det M = 1$, and 2 is a unit modulo any odd d . $\square$

We verified Theorem 6 by exhaustive search over all d^4 matrices for $2 \leq d \leq 12$; Table 1 records the census. Note that the obstruction is insensitive to primality: it holds at $d = 4, 8, 10, 12$ as well as at $d = 6$, and *fails* to obstruct at the odd composites $d = 9, 15$, where flat bilinear 2-unitaries do exist.

Corollary 7. *Any flat 2-unitary of order 36 — in particular any Butson-type AME(4, 6) representative — has a phase function that is not a bilinear form over $\mathbb{Z}/6$.*

This gives a precise sense in which the $d = 6$ solutions must be “non-Gaussian”: the quadratic-Gauss-sum mechanism that manufactures perfect tensors at every odd d is unavailable, and something genuinely different (in the known constructions, biunimodular vectors [8, 9]) must take its place.

5 Anatomy of two AME(4, 6) representatives

5.1 The golden representative

We parsed the analytic form of the golden AME(4, 6) matrix from the authors’ own source code and re-verified it independently. It has 112 nonzero entries with three amplitude values

$$a = \frac{1}{\sqrt{5+\sqrt{5}}}, \quad b = \frac{\sqrt{5+\sqrt{5}}}{\sqrt{20}}, \quad c = \frac{1}{\sqrt{2}},$$

d	parity	$\#\{M \text{ admissible}\}$	canonical solution
2	even	0	—
3	odd	8	$(1, 1; 1, 2)$ verified 2-unitary, flat
4	even	0	—
5	odd	192	$(1, 1; 1, 2)$ verified
6	even	0	—
7	odd	1080	$(1, 1; 1, 2)$ verified
8	even	0	—
9	odd	648	$(1, 1; 1, 2)$ verified
10	even	0	—
11	odd	9000	$(1, 1; 1, 2)$ verified
12	even	0	—

Table 1: Exhaustive census of bilinear phase forms $M \in M_2(\mathbb{Z}/d)$ yielding a flat 2-unitary of order d^2 . Existence holds exactly for odd d , in agreement with Theorem 6. Admissibility was tested by the criterion of Lemma 5 and confirmed numerically on representatives.

and phases in μ_{20} , $w = e^{i\pi/10}$. We confirm $b/a = \varphi$ exactly and $a^2 + b^2 = c^2 = \frac{1}{2}$; that U , U^R , U^Γ are unitary; and that all six bipartition purities equal $1/36$ to machine precision. The amplitude census is 40 entries of modulus a , 40 of modulus b , and 32 of modulus c .

Motivated by the exceptional 2-transitive action of $\mathrm{PGL}_2(\mathbb{F}_5) \supset \mathrm{PSL}_2(\mathbb{F}_5) \cong A_5$ on the six points of $\mathbb{P}^1(\mathbb{F}_5)$ — the source of the outer automorphism of S_6 , and of golden character values — it is natural to ask whether the golden state is covariant for a copy of A_5 acting on the six levels. We tested this exhaustively.

Proposition 8. *Let $\mathcal{E} \subset (\mathbb{Z}/6)^4$ be the support of the golden representative, colored by amplitude class. The group of quadruples $(\pi_1, \pi_2, \pi_3, \pi_4) \in S_6^4$ acting independently on the four legs and preserving $\mathcal{E}$ together with its coloring has order 2. Its nontrivial element is*

$$\pi_1 = (0\,1)(2\,3)(4\,5), \quad \pi_2 = (0\,5)(1\,4)(2\,3), \quad \pi_3 = \pi_4 = \mathrm{id}.$$

Moreover this element does not lift to a monomial symmetry: there are no diagonal unitaries D_L, D_R with $D_L(P_{\pi_1} \otimes P_{\pi_2}) U D_R = U$.

Proof (computational). Leg-wise color profiles reduce the search to 48, 48, 4, 8 candidates on the four legs; pair-profile filtering and constraint propagation on the fibers leave 2 quadruples, both of which are then verified to preserve all 112 colored edges. For the second claim, write $V = (P_{\pi_1} \otimes P_{\pi_2})U$; then $|V| = |U|$, and a monomial lift exists iff the phase ratio $\arg(V_{jk}/U_{jk})$ is of the separable form $r_j + c_k$ on the support. The support bipartite graph has 9 connected components, each of size 8; performing the propagation *within each component* (the assignment is free between components) yields residuals up to 2.51 radians, so no separable solution exists. $\square$

Corollary 9. *The golden representative is not covariant for any nontrivial group of level permutations acting monomially; in particular it carries no A_5 or $\mathrm{PGL}_2(\mathbb{F}_5)$ symmetry.*

Remark 10 (Scope). This is a statement about *this representative*. AME states are defined up to local unitaries, and a symmetry invisible in one frame may exist in another; the negative result rules out permutation-monomial symmetry in the analytic frame of [4], not in the whole LU class. We flag the possibility that the golden solution’s arithmetic (μ_{20}, φ) is a feature of the seed — the algorithm of [4] is initialized near the best classical permutation [14] — rather than of the state.

The next proposition explains where φ does come from, and is consistent with Proposition 8: the golden ratio is a *local* datum of the 2×2 blocks, not a shadow of a global icosahedral symmetry.

Proposition 11. *The three amplitudes of the golden representative satisfy the closed forms*

$$a^2 = \frac{1}{2\varphi\sqrt{5}}, \quad b^2 = \frac{\varphi}{2\sqrt{5}}, \quad c^2 = \frac{1}{2},$$

so that $a^2 + b^2 = c^2$ and $b/a = \varphi$. Equivalently, the within-block probability split is

$$\left(\frac{1}{\varphi\sqrt{5}}, \frac{\varphi}{\sqrt{5}}\right) = (0.2763932\dots, 0.7236067\dots),$$

which sums to 1 by the identity $\varphi^2 + 1 = \varphi\sqrt{5}$, and the normalized block vector is

$$\frac{(a, b)}{c} = \frac{(1, \varphi)}{\sqrt{2 + \varphi}} = (0.5257311\dots, 0.8506508\dots),$$

the pair of nonzero coordinates of a vertex of the regular icosahedron. Moreover this differs from the normalized column $(1/\varphi, 1/\sqrt{\varphi}) = (0.6180339\dots, 0.7861513\dots)$ of the Fibonacci F -matrix, which is normalized by the distinct golden identity $\varphi^{-1} + \varphi^{-2} = 1$.

Remark 12. Thus the golden amplitudes are *not* Fibonacci fusion data, and the two natural “golden splits” $(\varphi/\sqrt{5}, 1/(\varphi\sqrt{5}))$ and $(\varphi^{-1}, \varphi^{-2})$ must not be conflated. Combined with Proposition 8, the picture is: φ appears as the icosahedral vertex ratio inside each block, and does not propagate to a symmetry of the 36-officer array.

5.2 Block structure, and what the local datum actually is

The support of the golden representative is highly disconnected, and the components are the natural carriers of its arithmetic.

Proposition 13. *After row and column permutation the golden representative is block diagonal with exactly 9 blocks, each a unitary 4×4 matrix. Eight of them are mutually equivalent, with 12 nonzero entries and every row of squared moduli a permutation of*

$$(0, a^2, b^2, c^2) = (0, 0.1381966\dots, 0.3618033\dots, 0.5).$$

The ninth is exceptional: it has full support, 16 nonzero entries, and every row of squared moduli a permutation of (a^2, a^2, b^2, b^2) . Total support $8 \cdot 12 + 16 = 112$.

Proposition 14 (Support asymmetry). *The support of the golden representative is not symmetric under transposition. Its row and column support profiles are*

$$\text{rows: } 32 \times 3 + 4 \times 4, \quad \text{columns: } 16 \times 2 + 20 \times 4,$$

both summing to 112. The asymmetry is entirely accounted for by Proposition 13: each generic block has row supports $(3, 3, 3, 3)$ and column supports $(2, 2, 4, 4)$, contributing 16 columns of support 2; the exceptional block contributes four more columns of support 4. Moreover, reshaping each line as a 6×6 matrix, every one of the 36 columns has Schmidt rank exactly 2, while the rows have Schmidt rank 3 (32 of them) or 4 (4 of them).

Remark 15 (Which quantities are invariants). Several natural-looking quantities attached to a block are not well defined. The blocks of Proposition 13 are extracted by independent row and column index sets, so there is no canonical pairing of rows with columns; consequently the *eigenvalues* of a block are not invariants. Concretely, transposing two rows of the exceptional block changes its spectrum from $\{-0.979 - 0.206i, 0.486 - 0.874i, 0.604 + 0.797i, 0.929 - 0.371i\}$ to $\{-0.246 - 0.969i, -0.178 + 0.984i, 0.660 - 0.751i, 0.822 + 0.569i\}$. The quantities that are invariant, and which we use, are the singular values (all equal to 1, since each block is unitary) and the multiset of entry moduli, which is what Corollary 17 rests on.

Proposition 16. *The block amplitudes admit the parametrization*

$$a^2 = \frac{1}{4} \left(1 - \frac{1}{\sqrt{5}} \right), \quad b^2 = \frac{1}{4} \left(1 + \frac{1}{\sqrt{5}} \right),$$

so that, with $\theta = \arctan \varphi$ the block angle, $\cos 2\theta = -1/\sqrt{5}$ and $\sin 2\theta = 2/\sqrt{5}$. Hence the primitive datum of the block is $\sqrt{5}$, the quadratic Gauss sum of conductor 5, and $\varphi = b/a$ is derived from it by the half-angle identity $b/a = \sqrt{(\sqrt{5} + 1)/(\sqrt{5} - 1)} = \varphi$.

This settles, at the level of the 2×2 datum, the question of whether the golden block is Fibonacci fusion data.

Corollary 17. *No block of the golden representative is gauge-equivalent to the Fibonacci F -matrix.*

Proof. Gauge transformations act by $F \mapsto D_1 F D_2$ with D_i diagonal unitary, together with relabelling; the multiset of entry moduli is therefore a gauge invariant. The normalized golden datum has moduli $\{1/\sqrt{2+\varphi}, \varphi/\sqrt{2+\varphi}\} = \{0.5257311\dots, 0.8506508\dots\}$, while the Fibonacci F -matrix has moduli $\{1/\varphi, 1/\sqrt{\varphi}\} = \{0.6180339\dots, 0.7861513\dots\}$. The two multisets differ, and the underlying normalizations are the distinct golden identities $1 + \varphi^2 = \varphi\sqrt{5}$ and $\varphi^{-1} + \varphi^{-2} = 1$. $\square$

Remark 18. The broader question — whether some block is an F -matrix of *some* fusion category — is not settled by Corollary 17, but is now sharply constrained: any such category must reproduce the row profile $(0, a^2, b^2, c^2)$ or (a^2, a^2, b^2, b^2) . We also record a negative result guarding against an appealing guess: although the exceptional block's modulus pattern coincides with that of a Kronecker product $R \otimes H$ (R the 2×2 icosahedral rotation, H a flat 2×2 unitary), an exhaustive search over all $4! \times 4!$ row and column permutations shows that it does *not* factor as such a product once phases are taken into account. Matching modulus patterns do not imply tensor structure.

Remark 19 (Both families are quadratic Gauss sums at odd conductor). Proposition 16 places the two known families of $\text{AME}(4, 6)$ representatives in a common frame. The golden family has coefficient field $\mathbb{Q}(\zeta_{20})$, which contains the conductor-5 quadratic Gauss sum $\sum_a \left(\frac{a}{5}\right) \zeta_5^a = \sqrt{5}$; the Butson family of Proposition 21 has coefficient field $\mathbb{Q}(\zeta_6)$, which contains the conductor-3 Gauss sum $\sqrt{-3}$. Both are thus built over the quadratic Gauss sum of an odd prime conductor — the two smallest — while Theorem 6 says the even part of the modulus can supply no such construction at all. We state this as an observation about the two known solutions, not as a classification: by [6] the moduli space is infinite, and we have examined two points of it.

It is tempting to separate the golden and Butson representatives by their support density $F_1 := \#\{\text{nonzero entries}\}/1296$: one computes

$$F_1(\text{golden}) = \frac{112}{1296} = \frac{7}{81} = 0.0864\dots, \quad F_1(\text{Butson}) = 1.$$

This separation is real but must not be read as a statement about the $\text{AME}(4, 6)$ moduli space, because F_1 is not invariant under local unitaries: applying a generic $V_1 \otimes V_2 \otimes V_3 \otimes V_4$ to the golden state leaves the state's LU class unchanged while sending F_1 to 1. The corrected functional is the orbit minimum $F_1^*(\psi) := \min\{F_1(U) : U \in \text{LU-orbit of } \psi\}$. The following pins down its range.

Proposition 20. *Every 2-unitary of order 36 has at least 37 nonzero entries. Consequently*

$$\frac{1}{36} < F_1^* \leq \frac{7}{81}$$

for the golden state.

Proof. Suppose U is 2-unitary with exactly 36 nonzero entries. Since U is unitary every row and every column has norm 1, hence at least one nonzero entry; with only 36 entries available among 36 rows and 36 columns, each row and each column carries exactly one. So U is monomial, $U = DP$ with D diagonal unitary and P a permutation matrix; write $T_{jklm} \in \{0, 1\}$ for the 0/1 tensor of P , so that $|U_{jk;lm}| = T_{jklm}$.

Unitarity of U^R forces each of its rows to have norm 1: the row indexed by (j, l) has entries $U_{jk;lm}$ as (k, m) varies, so $\sum_{k,m} T_{jklm} = 1$ for every (j, l) . Likewise unitarity of U^Γ gives $\sum_{k,l} T_{jklm} = 1$ for every (j, m) , and unitarity of U gives $\sum_{l,m} T_{jklm} = 1$ for every (j, k) . These are exactly the conditions under which T encodes a pair of orthogonal Latin squares of order 6 [10]; the diagonal phases D play no role in the counting. No such pair exists [1, 3], a contradiction. $\square$

Whether the golden value $7/81$ is itself optimal — equivalently, whether 112 is the minimum support of any $\text{AME}(4, 6)$ representative — we leave open.

5.3 The Butson (Eisenstein) representatives

We rebuilt the biunimodular construction of [7, 8] from scratch: with F_6 the Fourier matrix, P the mod-6 generalized CNOT, and $K = (F_6 \otimes \mathbb{I})P(F_6 \otimes \mathbb{I})$, $L = (F_6 \otimes \mathbb{I})P(F_6^\dagger \otimes \mathbb{I})$, we form $\mathcal{U}_j = K \text{diag}(e^{2\pi i \Lambda_j}) L$. We verify $K = K^\dagger$, $L = L^\Gamma$, and that both are flat.

Proposition 21. $\mathcal{U}_1$ and $\mathcal{U}_2$ are flat 2-unitary matrices of order 36: all entries have modulus $1/6$, all of $\mathcal{U}$, $\mathcal{U}^R$, $\mathcal{U}^\Gamma$ are unitary, and all six bipartition purities equal $1/36$. Every entry is exactly a sixth root of unity: writing $\mathcal{U}_{jk} = \frac{1}{6} \zeta_6^{m_{jk}}$ with $m_{jk} \in \mathbb{Z}/6$, the reconstruction error is 8.3×10^{-15} . Hence $6\mathcal{U}_j \in \text{BH}(36, 6)$ and all phases lie in $\mathbb{Z}[\omega]^\times$.

For $\mathcal{U}_1$ the exponent census over $\mathbb{Z}/6$ is $(204, 180, 294, 180, 204, 234)$ for $m = 0, \dots, 5$. We then probed the structure of the phase function $m(x, y, u, v)$ on $(\mathbb{Z}/6)^4$.

Observation 22. For $\mathcal{U}_1$:

- (a) *Non-polynomiality.* m is not a polynomial of total degree ≤ 2 over $\mathbb{Z}/6$: fitting the 15 monomials of degree ≤ 2 in four variables against all 1296 points is inconsistent both modulo 2 (rank 11) and modulo 3 (rank 15).
- (b) *No translation covariance.* None of the shifts $(x, u) \mapsto (x+t, u+t)$, $(y, v) \mapsto (y+t, v+t)$, or a simultaneous shift of all four legs changes m by a constant, for any $t = 1, \dots, 5$.
- (c) *Distance from bilinearity.* The best pure bilinear form agrees with m on 372 of 1296 points (28.7%, against a 16.7% random baseline). The maximum is attained at more than one M (e.g. $M = (2, 3; 1, 2)$); only the value 372 is canonical.
- (d) *CRT non-factorization.* Write $\zeta_6^m = (-1)^m \bar{\omega}^m$, splitting the phase matrix into a χ_2 -layer $S \in \{\pm 1\}^{36 \times 36}$ and a χ_3 -layer $W \in \mu_3^{36 \times 36}$. Neither layer is unitary after normalization: the Gram spectrum of $S/6$ runs from 0 (so S is singular) to 9, and that of $W/6$ from $1/3$ to 7.

Part (d) is the structurally sharpest: the solution cannot be assembled from a real Hadamard factor and an Eisenstein factor separately. Together with Theorem 6 it says that the 2-adic and 3-adic layers of $\mathbb{Z}/6$ are irreducibly entangled in any flat order-36 solution — the same factorization $6 = 2 \cdot 3$ that destroys the field $\mathbb{F}_6$ and hence the bilinear route also forbids the solution from splitting along that factorization. Parts (a)–(c) confirm that the phase function is genuinely of “biunimodular” rather than Gauss-sum type.

Remark 23. Our transcription of the third vector Λ_3 from the published PDF failed its own biunimodularity test; we attribute this to text-extraction corruption on our side rather than to an error in [7], and have therefore excluded $\mathcal{U}_3$ from Proposition 21. The authors' machine-readable data is available at their repository.

6 A companion arithmetic thread: weight 36 and χ_{-3}

The objects of Sections 3–5 are governed by the modulus 6 and the field $\mathbb{Q}(\sqrt{-3})$. The same two data organize a classical arithmetic thread in which the number 36 reappears as a *weight* rather than a dimension. We record here what we could verify, and state plainly that we have no proven bridge between the two threads: the shared modulus is a thematic link, not a theorem.

Throughout, χ_{-3} denotes the odd quadratic character modulo 3, i.e. the Kronecker symbol $\left(\frac{-3}{\cdot}\right)$. This is the character in the Eisenstein master identity $r(n)/6 = \sum_{d|n} \chi_{-3}(d)$ for representations by $x^2 + xy + y^2$, and the non-principal character modulo 6 is induced by it; the two L -functions differ only by the Euler factor at 2, $L(s, \chi_6) = L(s, \chi_{-3})(1 + 2^{-s})$.

6.1 Degree of the modular-zero polynomial

Let $k = 12m$, so $\dim S_k = m$, and let $f_1, \dots, f_m$ be the normalized Hecke eigenforms. Define the *modular-zero polynomial* $P_k(X) = \prod_i \prod_\tau (X - j(\tau))$, the product over eigenforms and over their zeros τ in the open upper half-plane modulo $\mathrm{SL}_2(\mathbb{Z})$.

Proposition 24. *Generically* $\deg P_{12m} = m(m-1)$.

Proof. A normalized eigenform has $a_1 = 1$, hence $\mathrm{ord}_\infty f_i = 1$ exactly. The valence formula

$$\mathrm{ord}_\infty f + \frac{1}{2}\mathrm{ord}_i f + \frac{1}{3}\mathrm{ord}_\rho f + \sum_{P \neq i, \rho} \mathrm{ord}_P f = \frac{k}{12} = m$$

therefore leaves total order $m-1$ among the points of the upper half-plane. Away from the elliptic points these are $m-1$ zeros contributing $m-1$ distinct values of j ; summing over the m eigenforms gives $m(m-1)$. (Zeros at i or ρ , which carry weights $\frac{1}{2}$ and $\frac{1}{3}$, would alter the count; the statement is generic in that sense.) $\square$

The formula gives 0, 2, 6, 12, 20 for $m = 1, \dots, 5$. The case $m = 1$ is a consistency check: Δ has its only zero at the cusp, so $P_{12} = 1$. The value 6 at $m = 3$ matches the degree of the weight-36 polynomial obtained previously as a symbolic resultant. An independent computation reports irreducibility for $m = 2, 3, 4, 5$ and $\mathrm{sign}(\mathrm{disc}) = +, -, -, +$ respectively.

6.2 The weight-36 Hecke cubic

We computed S_{36} from scratch in the echelon basis $\Delta E_4^6, \Delta^2 E_4^3, \Delta^3$ (leading terms q, q^2, q^3) and applied $T_2 f = \sum_n (a(2n) + 2^{35} a(n/2)) q^n$.

Proposition 25. *In that basis*

$$T_2 = \begin{pmatrix} 1416 & 1 & 0 \\ 34629120000 & 144384 & -72 \\ -17915904000000 & -331776000 & -6144 \end{pmatrix},$$

with characteristic polynomial

$$\chi(x) = x^3 - 139656x^2 - 59208339456x - 1467625047588864.$$

This cubic is irreducible over $\mathbb{Q}$ and totally real, with roots $-165109.1678\dots$, $-26808.0076\dots$, $331573.1754\dots$, and

$$\text{disc } \chi = 2^{30} \cdot 3^{10} \cdot 5^2 \cdot 7^2 \cdot 23 \cdot 1259 \cdot \mathbf{269461929553},$$

which is not a square; hence $\text{Gal}(\chi/\mathbb{Q}) \cong S_3$ and the Hecke field is a totally real non-Galois cubic.

Remark 26 (Corroboration across independent computations). An independent computation communicated to us reports, for the degree-6 field $\mathbb{Q}(j_0)$ generated by a modular zero in weight 36: Galois group $C_2^3 \rtimes S_3 = 2 \wr S_3$ of order 48 (imprimitive, permuting three blocks of two — one conjugate pair per eigenform), signature $(r_1, r_2) = (4, 1)$, class number $h_K = 16$ with $\text{Cl}_K \cong C_2^4$, and field discriminant

$$-3 \cdot 5^2 \cdot 7^2 \cdot 23^2 \cdot 233 \cdot 269 \cdot 1259^2 \cdot \mathbf{269461929553}^2.$$

We did not reproduce the class-group data, having no class-group software available. We do note that our independently computed $\text{disc } \chi$ shares with it the primes 3, 5, 7, 23, 1259 and the 12-digit prime 269461929553, with the odd-exponent primes appearing squared upstairs — precisely the behaviour expected of a quadratic-type extension of the cubic subfield. Two computations performed by different routes agreeing on a 12-digit discriminant prime is strong mutual corroboration. Consistently, the negative discriminant matches signature $(4, 1)$ via $\text{sign} = (-1)^{r_2}$.

6.3 Unconditional numerics for $L(s, \chi_{-3})$

Writing $L(s, \chi_{-3}) = 3^{-s}(\zeta(s, \frac{1}{3}) - \zeta(s, \frac{2}{3}))$ and forming the Hardy function $Z(t) = e^{i\theta(t)}L(\frac{1}{2} + it, \chi_{-3})$ with $\theta(t) = \frac{t}{2} \log \frac{3}{\pi} + \Im \log \Gamma(\frac{3}{4} + \frac{it}{2})$, we find Z real to 10^{-21} , confirming root number +1.

Observation 27. $L(s, \chi_{-3})$ has exactly 34 zeros with $5 \leq t \leq 80$, the first eight at

$$8.0397, 11.2492, 15.7046, 18.2620, 20.4558, 24.0594, 26.5779, 28.2182.$$

At the first zero, $|L(\sigma + it_1)|$ is minimized at $\sigma = \frac{1}{2}$, where it equals 1.9×10^{-16} , against 2.97×10^{-2} already at $\sigma = 0.475$. The normalized Chebyshev bias satisfies

$$\max_{x \leq 50000} \frac{|\psi(x, \chi_{-3})|}{\sqrt{x}} = 1.0428 \quad \text{at } x = 269.$$

Lemma 28. *The normalized bias maximum is the same whether computed with the character modulo 3 or the character modulo 6.*

Proof. The two summatory functions differ only on prime powers 2^k . Up to $x = 269$ there are exactly eight such, 2, 4, $\dots$, 256, and $\chi_{-3}(2^k) = (-1)^k$, so their contributions cancel in pairs and $\psi(269, \chi_{-3}) = \psi(269, \chi_6)$. $\square$

Remark 29 (Honest limits). Observation 27 is evidence for GRH for this L -function up to an explicit height, and nothing more. A finite bound on $|\psi(x, \chi)|/\sqrt{x}$ certifies the Riemann hypothesis for $L(s, \chi)$ only up to a corresponding height; the implication “bounded normalized bias $\Rightarrow$ GRH” requires the bound for all x , and no such bound is proved here or elsewhere.

7 Discussion and open problems

A hexit holographic brick. A verified four-leg perfect tensor of local dimension six is exactly the ingredient required to build a HaPPY-type holographic code [13] on six-dimensional legs. Unlike the qubit pentagon code, such a brick cannot arise from a stabilizer/OLS construction, by Theorem 6 together with the nonexistence of OLS(6). Constructing the associated $((3, 6, 2))_6$ code explicitly and studying entanglement wedges on hexit legs is the natural next computation.

The MUB(6) wall. Theorem 3 exhibits, in dimension 3, an exact coincidence between an Eisenstein lattice structure and a complete MUB system. The question that this note raises but does not answer is whether the Butson representatives of Proposition 21 contain unbiasedness data in dimension six. A concrete program: decompose $6\mathcal{U}_1 \in \text{BH}(36, 6)$ into its 36 blocks of size 6×6 and test which pairs of blocks are unbiased. We regard Observation 22(d) as mild evidence *against* a clean outcome: an object that refuses to factor along $6 = 2 \cdot 3$ is unlikely to hand over a $\mathbb{Z}/6$ -structured set of bases. We state this as an open problem rather than a conjecture.

On the golden ratio. The existence of the Butson solutions shows that φ is not forced by $\text{AME}(4, 6)$: there are representatives whose entire coefficient field is $\mathbb{Q}(\zeta_6) = \mathbb{Q}(\sqrt{-3})$, arithmetically disjoint from the golden field $\mathbb{Q}(\zeta_{20})$ since $\mathbb{Q}(\zeta_{20}) \cap \mathbb{Q}(\zeta_6) = \mathbb{Q}$. Proposition 8 removes the most natural mechanism by which φ might have entered globally (icosahedral character values via $\mathbb{P}^1(\mathbb{F}_5)$), and Proposition 11 identifies where it does enter: the block-level relations $a^2 + b^2 = c^2$, $b/a = \varphi$, whose solution is the icosahedral vertex direction $(1, \varphi)$. The golden ratio in [4] is thus a local datum of a particular frame, plausibly inherited from the seeded numerical continuation, rather than an invariant of the $\text{AME}(4, 6)$ moduli space. A sharper form of the remaining question: is there a functional on that moduli space whose critical points are the golden and Butson representatives? Section 5.2 shows that the obvious candidate, support density, must first be replaced by its local-unitary orbit minimum F_1^* , for which Proposition 20 gives $\frac{1}{36} < F_1^* \leq \frac{7}{81}$.

The arithmetic thread. Section 6 is deliberately quarantined. Weight 36 and dimension 6 share the modulus and the field $\mathbb{Q}(\sqrt{-3})$, and the numeral 36 appears in both as $|\Phi^+(E_6)|$, as 6^2 , and as a weight; but we have no map between them and do not assert one. What the section does contribute is independent: a proof of the degree law $\deg P_{12m} = m(m-1)$ replacing an empirical observation, an independently computed Hecke cubic whose discriminant corroborates a separately obtained sextic field discriminant through a shared 12-digit prime, and reproducible unconditional numerics for $L(s, \chi_{-3})$. If a bridge exists, the most likely place to look is the appearance of $\mathbb{Q}(\sqrt{-3})$ on both sides: as the coefficient field of the Butson representatives, and as the CM field of the Eisenstein arithmetic.

A Status of claims

B Reproducibility

All results were obtained with NumPy in exact-integer or double precision as appropriate. Root systems were generated by reflection closure from the Cartan matrix; AME matrices were rebuilt from published defining data (Λ_j vectors, or the analytic entry list) and re-verified rather than trusted. Tolerances: 10^{-9} for unitarity and purity assertions; observed errors were at the 10^{-15} level. Exhaustive searches (Table 1, Proposition 8) enumerate their full stated domains.

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Claim	Status	Evidence
Lemma 5, Theorem 6 (parity obstruction)	proved	character-sum argument; exhaustive check $2 \leq d \leq 12$
Theorem 3 (a)–(d), E_6 MUB decomposition	verified computationally; underlying structure classical	explicit construction from Cartan matrix; full 12×12 overlap table
Proposition 21 ($\mathcal{U}_1, \mathcal{U}_2$ are Butson AME(4, 6))	verified (reproduces [7, 8])	rebuilt from F_6 and CNOT; purities and exact μ_6 entries
Proposition 8 ($\text{Aut} = \mathbb{Z}/2$, no monomial lift)	verified for this representative	exhaustive leg search; component-wise separability test
Proposition 11 (amplitude closed forms; $\neq$ Fibonacci data)	proved (exact identities)	closed forms checked symbolically and numerically
Proposition 13 (9 blocks: $8 + 1$)	verified computationally	support components; unitarity of each block
Proposition 14 (row/col support asymmetry; column Schmidt rank 2)	verified computationally	direct count; per-block reconciliation
Published entry list is exactly 2-unitary as published	verified	$\ U^\dagger U - I\ _F = 3.4 \times 10^{-15}$, likewise U^R, U^Γ
Proposition 16 (local datum is $\sqrt{5}$, φ derived)	proved	half-angle identities, $\cos 2\theta = -1/\sqrt{5}$
Corollary 17 (not Fibonacci F -data)	proved	modulus multiset is a gauge invariant
Exceptional block is <i>not</i> a Kronecker product	proved (exhaustive)	all $4! \times 4!$ permutations searched
Block is an F -matrix of <i>some</i> fusion category	open	constrained to the two row profiles
Proposition 20 ($F_1^* > 1/36$)	proved	monomial reduction + Tarry
F_1^* (golden) = $7/81$, i.e. 112 is minimal	open	only $F_1^* \leq 7/81$ is established
Observation 22 (a)–(d)	verified numerically	rank/consistency over $\mathbb{F}_2, \mathbb{F}_3$; shift tests; Gram spectra
Proposition 24 ($\deg P_{12m} = m(m-1)$)	proved (generic)	valence formula; $m = 1$ consistency check
Proposition 25 (weight-36 Hecke cubic, S_3 , totally real)	computed independently	T_2 built from q -expansions; disc factored
Sextic $\mathbb{Q}(j_0)$: $\text{Gal} = 2\wr S_3$, $h_K = 16$, $\text{Cl} \cong C_2^4$	reported independently; <i>not</i> reproduced here	corroborated only via shared discriminant primes
Observation 27 (34 zeros; bias 1.0428 at $x = 269$)	verified numerically	Hardy Z from Hurwitz zetas; direct ψ sum
GRH for $L(s, \chi_{-3})$	open	finite-height evidence only
Golden φ is an artifact of the seed	conjectural	supported by Prop. 8, 11 and existence of Butson solutions; not proved
A bridge between Sections 3–5 and 6	not claimed	shared modulus only
MUB(6) data inside $\mathcal{U}_1$	open	not tested here